

Similarity-Measured Isolation Forest: Anomaly Detection Method for Machine Monitoring Data

Changgen Li[✉], Liang Guo[✉], Hongli Gao[✉], and Yi Li[✉]

Abstract—A rough environment or unexpected accident of data acquisition instrument can introduce some anomalies in monitoring data. Those anomalies reduce data quality and lead to the incorrect recognition of machine health status. However, the research on anomaly detection of machine monitoring data (MMD) is very scarce. Moreover, anomaly detection methods in other fields cannot be directly applied to MMD. Therefore, a robust anomaly detection method called similarity-measured isolation forest (SM-iForest) is proposed to detect abnormal segments and the data therein. The inadaptability and instability of iForest were reduced while processing MMD benefiting from the characteristics of sliding-window processing. Moreover, an anomaly identification stage measuring the relative similarity of possible abnormal segments further improved the robustness of iForest. The effectiveness of the proposed method was verified with a vibration simulation signal and three sets of milling force signals. The results demonstrate that SM-iForest can detect the missing, shifting, and swelling segments robustly. Detection results of comparing seven methods suggest that SM-iForest is a promising method to detect MMD anomaly with a high detection rate and low false alarm rate.

Index Terms—Anomaly detection, cutting tool, data cleaning, isolation forest (iForest), machinery monitoring data (MMD).

NOMENCLATURE

Acronyms

AE	Autoencoder.
CM-PM	Condition monitoring to predictive maintenance.
CNC	Computer numerical control.
COPOD	Copula-based outlier detection.
CSMPAW	Continuous sequence of most possible abnormal window.
CSPAW	Continuous sequence of possible abnormal window.
FN	False negative.
FP	False positive.
iForest	Isolation forest.

Manuscript received October 15, 2020; revised January 7, 2021; accepted February 12, 2021. Date of publication March 9, 2021; date of current version March 17, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 51775452 and Grant 51905452, in part by the China Postdoctoral Science Foundation under Grant 2019M663549, and in part by the Local Development Foundation guided by the Central Government under Grant 2020ZYD012. The Associate Editor coordinating the review process was Dr. Arunava Naha. (Corresponding author: Hongli Gao.)

The authors are with the Engineering Research Center of Advanced Driving Energy-Saving Technology, Ministry of Education, Southwest Jiaotong University, Chengdu 610031, China, and also with the School of Mechanical Engineering, Southwest Jiaotong University, Chengdu 610031, China (e-mail: hongli_gao@swjtu.edu.cn).

Digital Object Identifier 10.1109/TIM.2021.3062684

GMM	Gaussian mixture model.
LODA	Lightweight on-line detector of anomalies.
LOF	Local outlier factor.
LSCP	Locally selective combination of parallel outlier ensembles.
PAW	Possible abnormal window.
PCA	Principal component analysis.
RUL	Remaining useful life.
SM-iForest	Similarity-measured isolation forest.
SR-CNN	A novel algorithm based on spectral residual and convolutional neural network.
SVDD	Support vector data description.
TP	True positive.

Notation

Ar_q	Data set(vector) that root node of an iTree contained.
d_w	Moving step size.
F_{Kcol}	The colth statistical feature of the k th data window.
F_{Kj}	The j th statistical feature of the k th data window.
F_i	The i th feature of possible-abnormal-data.
H_{max}	Height limitation of an iTree.
L_{avg}	Average path length that F_{Kj} has passed through in every iTree of an iForest.
L_i	Compensated path length in the i th iTree.
L_{org}	Length of raw vibration signal.
L_w	Width of sliding window.
Mat_{window}	Raw data matrix constructed by data segments (also called data windows).
P_{split}	Split value.
Sig_{org}	Raw vibration signal.
S_K, S_K^j	S_K is the value set of relative similarity, S_K^j is the ascending order of S_K .
W_i	The i th feature of standard-abnormal-data.
$w_a, w_b, \dots$	Possible abnormal window.
cl	Number of data segments from signals.
L	Path length that F_{Kj} has passed through in an iTree.
m	Total number of CSPAW.
ss	One side trimmed mean used to obtain relative threshold.
Score	Anomaly score of a data window.
SIM	Relative similarity between CSPAW and CSMPAW.
Th	Similarity threshold.

$\emptyset$	Length of vector Ar_q .
γ	Euler constant.
Φ	A set of abnormal data windows.
δ	The standard deviation of a set of data.
Δ	Detection accuracy of simulation signal.
$[t, A_0, f_n, \xi, T, f_s]$	Parameters used to construct periodic impulse of simulation signal.
$[\varphi, \beta, \varepsilon]$	Parameters used to construct additive drifting quantity of simulation signal.
$C(\emptyset)$	An adjustment of L .
$F \sim N(\mu, \sigma^2)$	Variable F obeys a normal distribution with a mean of μ and a variance of δ^2 .
$N_k(p)$	K-distance neighborhood of point p .
$lrd_k(p)$	Local reachability density of point p .

I. INTRODUCTION

WITH the advancement in sensor and testing technology, it is easier to collect monitoring data from machines. Collection of monitoring data can more efficiently diagnose machine faults and predict more accurately its RUL. Additionally, the diagnoses and prediction results can be used for predictive maintenance of mechanical equipment. The detailed CM-PM framework is illustrated in Fig. 1. This framework can significantly reduce maintenance costs of essential components and improve the efficiency and security of machines [1]–[6]. The CM-PM framework is composed of three modules: monitoring, diagnosis, and prognosis. Monitoring is fundamental to diagnosis and prognosis. In this module, the original monitoring information, such as vibration signals, is collected by sensors and used to detect the incipient fault. If a fault occurs, the diagnosis module can identify its pattern and severity to locate the source and take decisions regarding maintenance. A health indicator can be constructed for the prognosis module owing to historical machine degradation knowledge acquired with the monitoring and diagnosis module. The RUL of the machine component can be estimated using an indicator to provide maintenance suggestions before losing efficacy.

However, the quality of the raw data that are fed into the CM-PM framework is rarely perfect because most machines work in very rough environments; random disturbances in such environments can impact the acquisition or transmission of monitoring data and generate its abnormal segments [7]–[9]. These segments can severely weaken the accuracy and completeness of the data and lead to poor-quality in data extraction [10]. Poor-quality data pose a significant challenge in extracting useful knowledge for the diagnosis and prognosis module. Furthermore, it may lead to misjudging the state-of-health and RUL of a machine [11], [12]. Therefore, it is necessary to improve monitoring data quality before feeding them into condition monitoring models.

In this article, we consider abnormal data segments and the data therein as anomalies. These anomalies deviate from the expected value and often have various appearances, such as shifting and missing. Fig. 2 shows a set of cutting force signals collected from a CNC machining center while processing a metal part. These anomalies indicate their distinctive formation mechanism compared to normal data [13]. More-

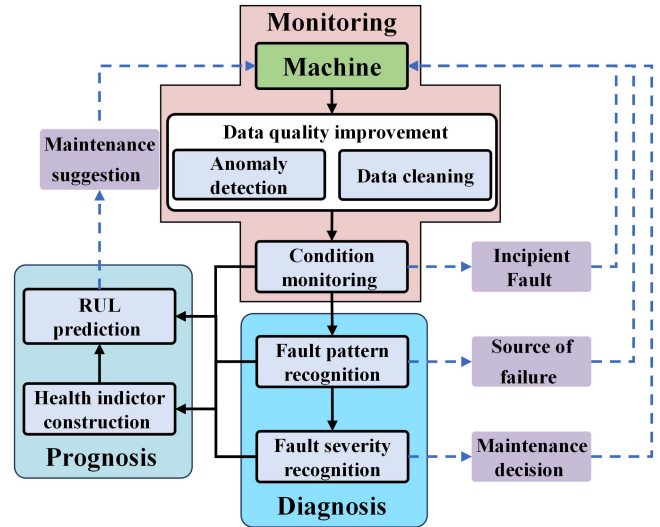


Fig. 1. CM-PM framework of the machine.

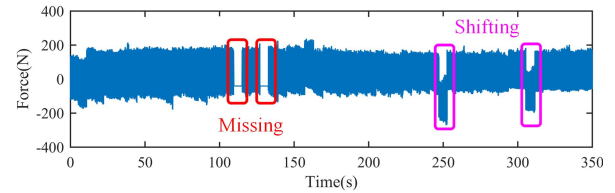


Fig. 2. Abnormal segments in monitoring data.

over, these anomalies cannot be considered as contextually attributable [14]. Therefore, if these anomalies are cleaned or replaced by data whose characteristics are contextually attributable, the quality of monitoring data would increase significantly. Moreover, detecting anomaly in raw monitoring data is an indispensable process before cleaning and replacing them.

The anomaly detection methods are generally categorized into three operating modes: supervised, semisupervised, and unsupervised [15]. The supervised methods require both labeled normal and abnormal samples to train a detection model.

Therefore, such methods are often preferred when sufficient labeled data are available. However, the semisupervised methods only need data from either of the two samples or a small number of data containing both the samples to train the detection model [16]. On the contrary, the unsupervised anomaly detection method does not require any labeled data to train a detection model, making it adaptive [17]. Unlabeled MMD is more common in most cases, making the unsupervised method a more feasible option for MMD anomaly detection.

Various studies regarding supervised anomaly detection methods have been published. Simon Hawkins *et al.* [18] constructed a neural network using multilayer perceptron. This neural network had equal amounts of output and input neurons, which is useful in measuring anomaly scores through the reconstruction error of each input data. The neural network-based method can isolate abnormal data; however, issues, such as over-fitting, can potentially arise. Additionally, research on semisupervised anomaly detection was also given considerable attention. It includes notable methods, such as GMM [19], AE [20], and SVDD [21].

Moreover, various unsupervised methods have also been proposed. Guo *et al.* [12] used the 3σ rule to detect the trend burrs of the health indicator. Hardin and Rocke [22] proposed a method based on the minimum covariance determinant estimator to locate anomalies in a multiple cluster setting. The Grubbs test—a classical anomaly detection method—was proposed by Grubbs [23] to detect anomalies in sets of univariate samples. Though such statistical methods [12], [22], [23] enable fast calculations, their performance may not be satisfactory when processing data with unknown distribution [24]. Shyu and Chen [25] and Lakhina *et al.* [26] demonstrated that the methods based on PCA could efficiently detect anomalies, which is a bit different method than those mentioned above. Yuen and Ortiz [27] improved an anomaly detection algorithm for linear regression involving correlated data. Jordaan and Smits [28] proposed an anomaly detection method based on a support vector machine. These four models [25]–[28] can only detect anomalies accurately when the data are correlated. Therefore, these methods are unfeasible when processing data without global correlation.

Additionally, the proximity-based method is also a representative category of unsupervised methods, which includes clustering-based methods, distance-based methods, density-based methods, and iForest. The clustering-based method [29] calculates the distance between each instance and the gravity center of n major categories to decide if the instance can be categorized as an anomaly. The k -nearest neighbors-based method [30] is a typical distance-based anomaly detection method, requiring a prespecified k and a large amount of distance calculation for each instance. Therefore, it is unsuitable for detecting anomalies in big data. The LOF-based method [31] is a typical density-based anomaly detection method. However, it is unsuitable for MMD containing numerous anomalies in one abnormal data segment owing to its local calculation characteristics that ignore global features. Therefore, Xu *et al.* [32] proposed an improved LOF anomaly detection method to detect abnormal data from mechanical vibrations. Several cases were used to demonstrate its accuracy and efficiency. Liu *et al.* [33] proposed iForest, which is suitable for distributed computation because all the iTrees are constructed independently. Moreover, its time complexity is $O(n)$, which means it is a linear-time algorithm; therefore, it is suitable for anomaly detection in big data. However, iForest is not a perfect method despite its benefits. For example, iForest is only applicable to data sets with a few anomalies and unsuitable for high-dimensional data sets.

Furthermore, it is important to note that various anomaly detection methods were proposed for different purposes. However, almost no method paid attention to MMD, largely due to some difference in characteristics between machine monitoring signals and various signals across other fields. For example, the sharp deviation of a single data point from the dense cluster is universal in certain fields. However, the deviation of certain data segments is more common in MMD. These abnormal segments can be considered as global anomalies. Therefore, local anomalies have a negligible influence on MMD than global anomalies. Moreover, most machine learning-based diagnosis and prognosis models can weaken the influence of local anomalies. Furthermore, it can be established

by analyzing the dynamical model of rotary machinery that periodicity is a fundamental attribute of mechanical monitoring signals [34]. Moreover, periodical components can also exist in MMD when faults occur in machinery. These conditions make duplicate value in MMD belong to normal data. However, in the other fields of research, duplicate values are often treated as anomalous data [35].

On the other hand, many anomaly detection methods used in other fields are also task-oriented. For example, Zou *et al.* [36] proposed an optimized iForest algorithm focusing on cloud computing, which sets a weight value for each of the four resource indicators: CPU usage, memory usage, disk read and write rates, and network speed. This method was designed to calculate the container abnormality to improve the security and stability of the cloud data container. Therefore, it is inapplicable to MMD anomaly detection because it does not consider the timing relationship among the containers. Tao *et al.* [37] proposed a parallel algorithm based on iForest and Spark to detect network traffic anomaly because the existing single-machine-oriented method cannot store and process large-scale network traffic data. The method assigned data processing tasks to multiple computing nodes to improve calculation efficiency. However, it only changed the way data were processed, leaving the data processing flow unimproved. Moreover, it was proposed to detect anomalies in large-scale network traffic data. Therefore, it is not suitable for MMD anomaly detection.

It is challenging to detect anomalies in MMD by directly employing methods from other fields. Therefore, iForest was improved to detect anomalies in MMD. The experimental results indicate that the proposed method can detect anomalies robustly and accurately.

The key insights and contributions of this letter are summarized as follows.

- 1) An anomaly detection method for MMD—SM-iForest—is proposed. First, a sliding window was utilized to divide raw data into various data segments. Second, sampled data sets from a feature matrix were used to train an iForest. Thereafter, the feature matrix was reused to measure the anomaly score of each data segment. Third, an anomaly identification stage that measures the similarity of possible abnormal segments was used to identify abnormal data segments based on anomaly scores. Consequently, data located in these abnormal segments were classified as abnormal data.
- 2) This article presents a feasible and general process to detect anomalies in MMD. It starts with a data preprocessing stage, followed by an anomaly score estimation stage, and ends with an anomaly recognition stage. This process flow provides a framework reference for readers who want to detect MMD anomalies by other unsupervised algorithms.

The rest of the letter is arranged as follows: Section II details the methodology of SM-iForest. Section III presents simulation data that were used to verify the effectiveness of the proposed method. Section IV demonstrates the performance of SM-iForest using monitoring data sets collected from a CNC machining center. Finally, the conclusions are drawn in Section V.

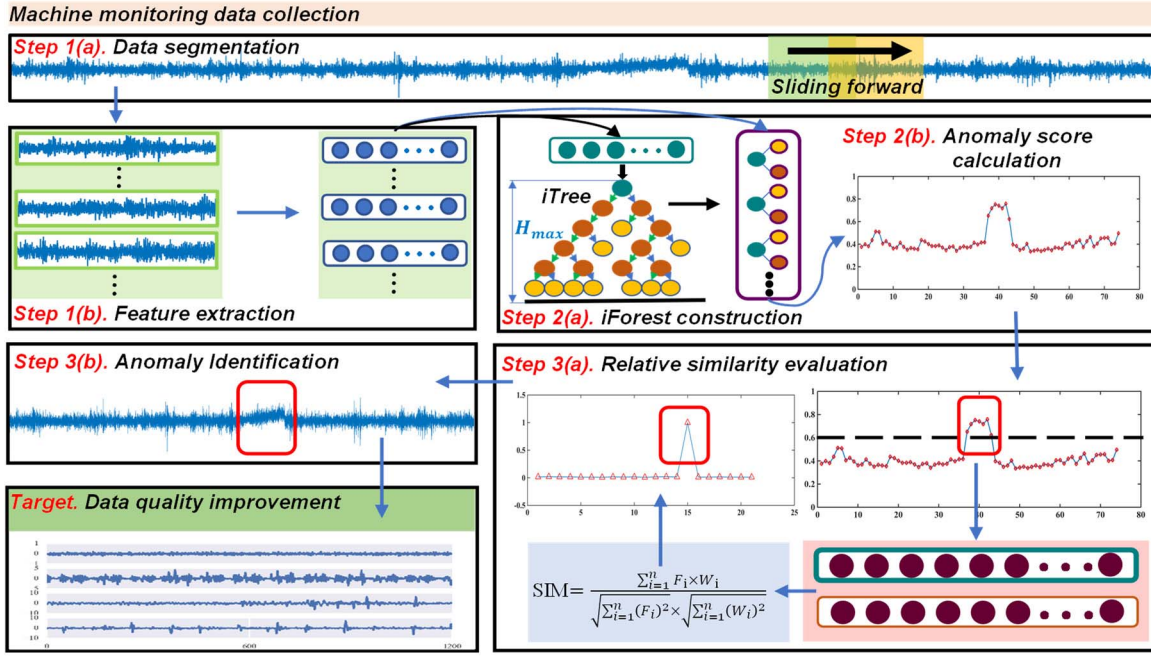


Fig. 3. Flowchart of the proposed method.

II. PROPOSED METHOD

The flowchart of the proposed method is illustrated in Fig. 3. First, collected data were divided into a series of data segments, also called data windows in this article, using a rectangle sliding window. After that, various numerical features, such as root mean square and kurtosis of each data window, were extracted. Second, these features were used to train a number of independent iTrees—the main component of an iForest. These features were then fed into the iForest to calculate the anomaly score of each data window. Third, the relative similarities between possible abnormal segments were inspected. Those that satisfy a similarity criterion were classified as abnormal windows. Moreover, abnormal data located in these abnormal windows were recognized. The located anomalies were then cleaned and replaced with context-sensitive data to improve data quality. Further details of the proposed method are described in the following three subsections.

A. Construction of Multidomain Feature Matrix

First, a raw data matrix was constructed from the data located in data windows. These data windows were obtained by a sliding window, which divides the original time-domain signal into a series of data segments, as shown in step 1(a) of Fig. 3. The overlapping fraction of two adjacent sliding windows was set at 90% [32]. For example, if a vibration signal, Sig_{org} , has the original length L_{org} . The width of the sliding window was set at L_w and the moving step size was set at d_w . Then, the vibration signal can be split into cl data segments to construct an $L_w \times cl$ raw data matrix named Mat_{window} , where

$$cl = \lceil (L_{org} - L_w) / d_w \rceil + 1. \quad (1)$$

In this example, it should be noted that the overlapping fraction of two adjacent sliding windows was set by controlling d_w .

Specifically, the raw data matrix was constructed by moving a sliding window from left to right on a raw signal. The right edge of the sliding window was named the “head of the sliding window,” and the right edge of the raw signal was called the “tail of the raw signal.” The segmentation would stop after the head of the sliding window exceeded the tail of the raw signal. Thereafter, the data read by the sliding window in every step were stored in a data window. Please note that the status of the last data window might be a little complex because the head of the sliding window and tail of the raw signal might not meet in the last moving step. Therefore, we filled the excess part of the last sliding window with 0, and it was excluded from the calculation of the statistical features. The specific procedures of data segmentation and raw data matrix construction are described in Algorithms 1 named “signal_Segment.”

Time-domain signals are the raw materials of machinery RUL prediction and fault diagnosis, which contain copious information; however, only a small percentage of them are useful. Moreover, using huge amounts of raw signals for diagnosis would expend a considerable percentage of memory space of the processor and reduce the efficiency of data processing [38]. This shortcoming is also faced when using raw signals for anomaly detection. Therefore, it is necessary to extract the statistical features for reducing the interfering signals present in raw data for efficient anomaly detection. The proposed method extracts the time domain, frequency domain, and wavelet packet features of each data window to construct a multidomain feature matrix, $Mat_{features}$, as given in

$$Mat_{features} = \begin{bmatrix} F_{11} & \cdots & F_{1\ col} \\ \vdots & \vdots & \vdots \\ F_{K1} & \cdots & F_{K\ col} \\ \vdots & \vdots & \vdots \\ F_{c1} & \cdots & F_{c\ col} \end{bmatrix} \quad (2)$$

Algorithm 1 Signal_Segment($\text{Sig}_{\text{org}}, L_w, d_w$)

Inputs: Sig_{org} -original time-domain signal, L_w -width of sliding window, d_w -moving step length of sliding window

Outputs: $\text{Mat}_{\text{window}}$ -data window matrix

```

1: Calculate cl
2: Initialize  $\text{Mat}_{\text{window}}$ 
3: for  $i = 1$  to cl do
4:   if  $i = 1$ 
5:     Assign the data from 1 to  $L_w$  in  $\text{Sig}_{\text{org}}$  to the  $i^{\text{th}}$  column
       of the  $\text{Mat}_{\text{window}}$ .
6:   else if  $i! = \text{cl}$ 
7:     Assign the data who are located from the  $(i \times d_w + 1)^{\text{th}}$ 
       to the  $(i \times d_w + 1 + L_w)^{\text{th}}$  in  $\text{Sig}_{\text{org}}$  to the  $i^{\text{th}}$  column
       of  $\text{Mat}_{\text{window}}$ .
8:   else
9:     Assign the data who are located from the  $(i \times d_w + 1)^{\text{th}}$ 
       to the end of  $\text{Sig}_{\text{org}}$  to the  $i^{\text{th}}$  column of  $\text{Mat}_{\text{window}}$  and
       replace the Null in the  $i^{\text{th}}$  column with 0.
11: End if
12: End for

```

where $[F_{K1}, F_{K2}, \dots, F_{K\text{col}}]$ is the statistical feature vector of the K th ($K = 1, 2, 3, \dots, \text{cl}$) data window. In this vector, col denotes the total quantity of types of extracted features.

The complexity of monitoring data made it challenging to identify the difference in features between abnormal segments and normal segments. Moreover, the dissimilarity between certain data segments might be negligible when considering only one type of feature. However, when all the features were considered, the differences between data segments were significant. Therefore, if the features are improperly selected or the feature types are insufficient, the detection accuracy can greatly reduce [39]. Therefore, all the features of the constructed feature matrix were employed to estimate the degree of abnormality of each data window to achieve a better detection result. Additionally, the extracted features were normalized based on the feature category, i.e., we normalized each column of the feature matrix, $\text{Mat}_{\text{features}}$, before using it to estimate anomaly score. If the normalization is not applied, the features with a high numerical value can significantly weaken the effect of features with a low numerical value.

B. Anomaly Score Estimation

The abnormal score of each data window was estimated using the iForest algorithm after constructing a multidomain feature matrix. This algorithm consisted of two modules: iForest construction and abnormal degree assessment. iForest was constructed using iTrees. The schematic diagram of one iTTree is shown under Step 2(a) *iForest construction* of Fig. 3. The dark green node of the schematic is called the root node, which contains a data set obtained by sampling from a column of the multidomain feature matrix. Thereafter, data smaller than a split value are stored in the left subtree node; the split is marked by a green arrow. The split value was randomly selected, such that it was larger than the minimum but smaller than the maximum of the data set. Similarly, data larger than

the split value are stored in the right subtree node, marked by a blue arrow. The yellow nodes are external nodes, i.e., they do not need a next level split. The height of iTTree is limited to the black line in this schematic. The constructed iForest and the feature matrix were applied in the abnormal degree assessment module to calculate the anomaly score of each data window. Details of the two modules are described below.

Module 1: iForest construction.

Step 1: Randomly select a type of feature from the features that are used to construct the feature matrix, i.e., the j th ($j = 1, 2, 3, \dots, \text{col}$) column in $\text{Mat}_{\text{features}}$.

Step 2: The data under this feature are randomly sampled to form the data set of the root node, i.e., sampling from data in the j th column of $\text{Mat}_{\text{features}}$ to form a data set Ar_q that the root node of an iTTree contained.

Step 3: Randomly select a split point P_{split} from max and min values of data set of the node.

Step 4: Compare all the elements in the node with the split value, P_{split} . The elements smaller than P_{split} are stored in the left subtree node, while the elements larger than P_{split} are stored in the right subtree node.

Step 5: Repeat steps 3 and 4 for the left and right subtree node until 1) only one data is left in the node, 2) all data in the node are of equal value, or 3) the iTTree reaches height limitation. The height limitation, H_{max} , of an iTTree is calculated as

$$H_{\text{max}} = \lceil \log_2 \emptyset \rceil \quad (3)$$

where $\emptyset = |\text{Ar}_q|$. The height of an iTTree is the largest number of edges in a path from the root node to a leaf node. The height limitation of iTTree is set because the data points that have a shorter path length than the height limitation are more likely to be anomalous.

These five steps constitute an iTTree construction algorithm. An iForest consists of multiple independent iTrees. Therefore, its construction algorithm can be divided into three steps. First, initialize the number of iTTree, n_{iT} . Second, construct an iTTree using the iTTree construction algorithm. Third, repeat the second step until the number of iTrees reaches the limitation initialized in the first step.

Module 2: Abnormal degree assessment. This module describes how to calculate the abnormal score of a data window represented by multidomain features using iForest.

Step 1: Get the feature marked by the i th iTTree, i.e., the j described in iTTree construction algorithm.

Step 2: Obtain F_{Kj} ($K = 1$ or $2, \dots$, or cl) from the feature matrix, $\text{Mat}_{\text{features}}$.

Step 3: Access the split value, P_{split} , from the node. If $F_{Kj} < P_{\text{split}}$, compare it with the split value of the left subtree node. If $F_{Kj} > P_{\text{split}}$, compare it with the split value of the right subtree node.

Step 4: Repeat step 3 until the node is an external node, and return the path length, L , that F_{Kj} passed through in the current tree, i.e., the number of times that F_{Kj} was compared with P_{split} .

Step 5: Compensate L as

$$L_i = L + C(\emptyset) \quad (4)$$

where L_i denotes the compensated path length in the i th iTree. $C(\emptyset)$ is an adjustment given in

$$C(\emptyset) = 2[\ln(\emptyset - 1) + \gamma] - \frac{2(\emptyset - 1)}{\emptyset} \quad (5)$$

where γ is the Euler constant.

Step 6: Repeat steps 1–5 on the $(i + 1)$ th iTree until the compensated path length, L_i , of F_{Kj} in all the iTrees of the iForest is obtained.

Step 7: Average the compensated path length as follows:

$$L_{\text{avg}} = \frac{\sum_{i=1}^{n_{iT}} L_i}{n_{iT}}. \quad (6)$$

Step 8: Calculate the anomaly score of the data window according to

$$\text{Score} = 2^{-\frac{L_{\text{avg}}}{C(\emptyset)}}. \quad (7)$$

C. Anomaly Identification

The position of anomalies was found after obtaining the anomaly score of each data window. In the proposed method, the threshold of the anomaly score was set at 0.6 [33], which means that the data window with a score exceeding 0.6 can be a PAW. Furthermore, the special process of dividing the raw data in which the i th sliding window partially overlaps with the $(i + 1)$ th window constructing the PAW could appear continuously. Therefore, we consider these continuous PAWs as an extended PAW. Moreover, continuous PAWs as a whole or independent PAW was marked as a CSPAW Data located in each CSPAW were possible abnormal data. For example, we obtained a set of abnormal data windows, Φ , as follows:

$$\Phi = \begin{bmatrix} w_a & w_b & w_c \\ w_d & w_{d+1} & w_{d+2} \\ w_e & w_f & w_g \\ w_h & w_{h+1} & w_{h+2} \end{bmatrix} \quad (8)$$

where w denotes a PAW. The continuous PAWs in Φ are $[w_d, w_{d+1}, w_{d+2}]$ and $[w_h, w_{h+1}, w_{h+2}]$. Hence, the CSPAWs include $w_a, w_b, w_c, [w_d, w_{d+1}, w_{d+2}], w_e, w_f, w_g$, and $[w_h, w_{h+1}, w_{h+2}]$.

The window in CSPAW with the highest anomaly score was considered to be a part of a CSMPAW. The data existing in CSMPAW were marked as standard abnormal data. Specifically, if w_{d+1} is the window with the highest anomaly score in Φ , the CSPAW $[w_d, w_{d+1}, w_{d+2}]$ containing that window is categorized as the CSMPAW. Accordingly, data in $[w_d, w_{d+1}, w_{d+2}]$ were standard abnormal data.

Based on the background information mentioned above, the first step to identify the anomaly was to obtain CSPAW, CSMPAW, possible abnormal data, and standard abnormal data.

Thereafter, the features of possible abnormal data and standard abnormal data were extracted to calculate the similarity between each CSPAW and CSMPAW.

Finally, CSPAW with a similarity greater than a threshold were marked as abnormal windows. Consequently, data existing in these data windows are abnormal data.

Specifically, possible abnormal data of a CSPAW have statistical features, $F_i (i = 1.2, \dots, n)$, and the standard abnormal

data have statistical features, $W_i (i = 1.2, \dots, n)$. Relative similarity, SIM, of the CSPAW and CSMPAW is calculated by

$$\text{SIM} = \frac{\sum_{i=1}^n F_i \times W_i}{\sqrt{\sum_{i=1}^n (F_i)^2} \times \sqrt{\sum_{i=1}^n (W_i)^2}}. \quad (9)$$

CSPAW, whose SIM is greater than the similarity threshold, Th , is an abnormal window. Th includes two parts: absolute threshold set at 0.99 and relative threshold dynamically set using (10). In the anomaly identification stage, the similarity between CSMPAW and CSPAW was first compared to the absolute threshold. If the similarity was larger than the absolute threshold, the CSPAW was directly classified as an abnormal windows. Otherwise, the similarity was compared to the relative threshold. After that, CSPAW with a similarity larger than a relative threshold was classified as abnormal windows.

$$Th = ss + (1 - st) \times (\text{MAX}(S_K) - ss) \quad (10)$$

where st is a detection sensitivity within (0.1) range. The higher the value, the more sensitive detection is to anomaly scores. Furthermore, ss represents the one-side-trimmed mean of all the similarities, which is expressed as:

$$ss = \frac{\sum_{j=\lceil m \times 3\gamma \rceil}^m S_K^j}{m - \lceil m \times 3\delta \rceil} \quad (11)$$

where m is the total number of CSPAW obtained in the first step of anomaly identification. $S_K (k = 1, 2, \dots, m)$ denotes the value set of relative similarities between each CSPAW and CSMPAW. Furthermore, $S_K^j (j = 1, 2, \dots, m)$ is the ascending order of S_K . δ represents the standard deviation of all the similarities, whose expression is is given in (12).

Finally, the data from abnormal windows were identified as abnormal data.

$$\delta = \sqrt{\frac{1}{m-1} \sum_{k=1}^m \left(S_K - \frac{\sum_{k=1}^m S_K}{m} \right)^2}. \quad (12)$$

III. CASE STUDY I: FAULTY BEARING SIMULATION SIGNAL

In this section, a simulation data set of faulty bearing vibration signals with a drifting segment was used to verify the effectiveness of the proposed MMD anomaly detection method.

A. Data Description

In general, data collected from a fault bearing exhibit some common characteristics, and the periodic impulse is a notable one. These periodic impulses are generated by recurrent collisions between a defect and mating surface [40]. Therefore, the periodic impulse used in this case is described as

$$S(t) = A_0 e^{-2\pi f_n \xi} \sin(2\pi f_n \sqrt{1 - \xi^2} t). \quad (13)$$

The specific parameters are given in Table I. Additional information to construct the simulation signal, including

TABLE I
PARAMETERS OF SIMULATION SIGNAL

Symbol	Name	Value
t	Duration of one impulse	Independent variable
A_0	Shifting constant	2
f_n	Natural frequency	3000Hz
ξ	Damping factor	0.1
f_s	Sampling frequency	20kHz
T	Impulse period	0.04s
—	Number of sampling points	1×10^6
—	SNR of Gaussian white noise	10 dB
φ	—	3
β	—	0.2
ε	—	0.8

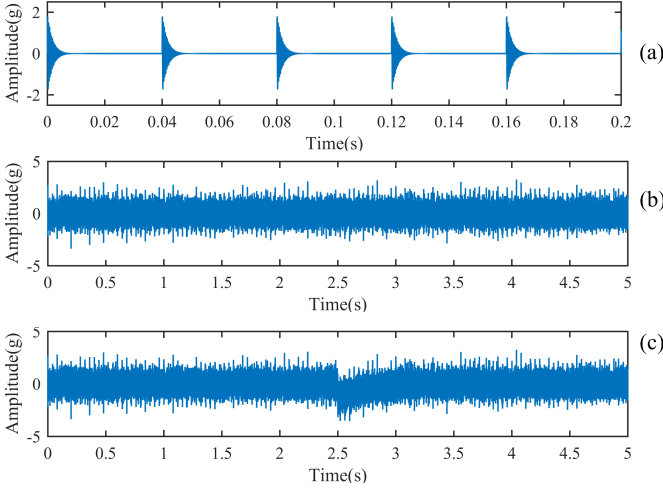


Fig. 4. Construction of simulation signal. (a) Zoomed-in view of periodic impulses. (b) Periodic impulses signal with Gaussian white noise. (c) Simulation signal with shifting segment.

impulse period, sampling frequency, and the number of sampling points, is also presented in Table I. The zoom-in view of these impulses is shown in Fig. 4(a). After that, Gaussian white noise with the SNR of 10 dB was added to the periodic impulses, as shown in Fig. 4(b). Finally, the simulation signal containing a shifting segment—a common abnormal data segment—was produced by adding the signal from $t_1 = 2.5$ s to $t_2 = 3$ s with an additive shifting quantity, S_1 .

$$S_1(x) = -\varphi e^{-(\beta x + \varepsilon)}. \quad (14)$$

The specific parameters are given in Table I. x is made up of $(1/2)f_s$ pieces of data, which are sampled evenly from the closed interval $[1, 10]$. Moreover, the simulation signal is shown in Fig. 4(c).

B. Anomaly Detection for Simulation Signal

Usually, abnormal data exhibit patterns that are different from normal data. The proposed method is used to detect these abnormal data. The software and hardware configuration used to process these data sets are presented in Table II.

1) *Detection Process of Simulation Signal Anomaly*: First, the whole signal was divided into 325 data windows using a sliding window by manually setting the window length, $L_w = 3000$, and 90% overlapping adjacent windows [32]; it means that the window moving step length was $d_w = 300$. Therefore, the simulation signal containing 10^6 data points

TABLE II
SOFTWARE AND HARDWARE CONFIGURATION

Category	Name	Version
Software	MATLAB	R2014b
Hardware	CPU	AMD Ryzen 7 2700X 3.70 GHz
Hardware	GPU	Nvidia GeForce GTX 1660 SUPER 6GB
Hardware	RAM	Kingston DDR4 16 GB 2666MHz

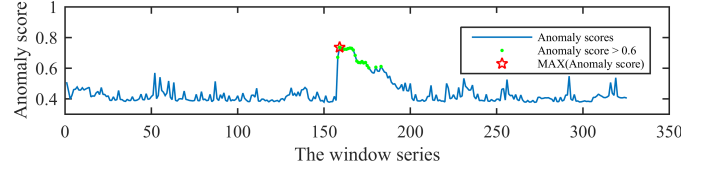


Fig. 5. Anomaly score of each window.

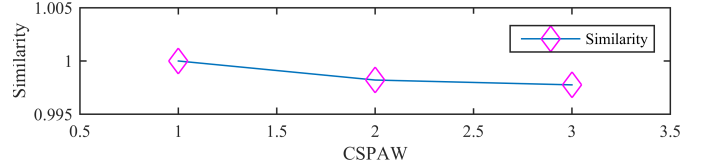


Fig. 6. Relative similarity between CSPAW and CSMPAW.

TABLE III
DETECTION RESULT OF RELATIVE AND HYBRID THRESHOLD

	Start time(s)	End time(s)	Accuracy Δ
Relative threshold	2.3699	2.8050	61%
Hybrid threshold	2.3699	2.9100	82%

was split to construct the window matrix of raw data with a size of 3000×325 . The last column of the matrix contained 2800 elements from the signal, and the rest was complemented by 0. After that, the statistical features, including the time domain, frequency domain, and wavelet packet of each column in the matrix, were extracted to construct a 325×29 feature matrix.

Moreover, the anomaly score of each window was calculated using iForest. The result is shown in Fig. 5. In this case study, the number of iForest was set to 10 to ensure the robustness of the estimation. Each forest contained 100 iTrees [33], whose height was limited to $\lceil \log_2 256 \rceil$. There were some windows that could be regarded as PAWs because their anomaly score was greater than 0.6.

Furthermore, the relative similarities between CSPAW and CSMPAW were calculated. The result is shown in Fig. 6. We find that there are three CSPAW, and the similarity of each is larger than 0.99. The high similarity means that all three CSPAW have indistinguishable features compared to CSMPAW. If we persistently use the relative threshold to identify abnormal CSPAW, there can be some loss in identification accuracy. Therefore, a hybrid threshold would be a better choice. The relative and hybrid threshold can recognize the location where the drifting segment generates, but the hybrid threshold provides a more accurate detection result than the relative threshold, as shown in Table III.

2) *Results and Discussion*: The detection result is visualized in Fig. 7. The red points are abnormal data. The simulation result suggested that the proposed method detected abnormal data in the shifting segment at an accuracy of 82%. The accuracy, Δ , is calculated by (15), where t_3 , t_4 represent the detected start time and end time of the abnormal segment,

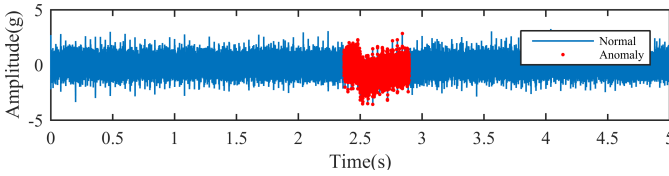


Fig. 7. Visualization of detection results of the simulation signal.

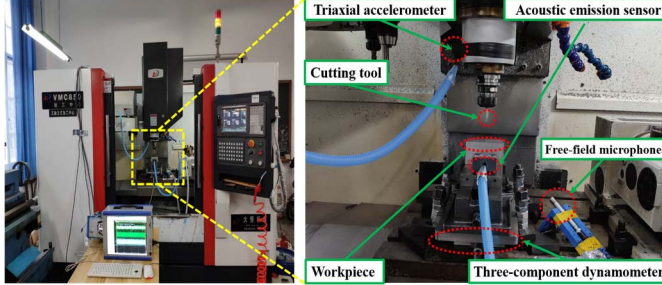


Fig. 8. Experimental condition.

respectively. A small abnormal data close to 3 s are challenging to be identified in one detection owing to their relatively small shifting quantity. However, their influence is negligible. Moreover, they can be detected by cleaning the detected data from the signal and again executing the method.

$$\Delta = \frac{\min(t_2, t_4) - \max(t_3, t_1)}{t_2 - t_1} \times 100\%. \quad (15)$$

IV. CASE STUDY II: MILLING FORCE SIGNAL

To further verify the effectiveness and superiority of SM-iForest, it was used to detect three kinds of force signal anomalies. Moreover, it was also compared to seven published detection methods.

A. Experimental Description

As shown in Fig. 8, the experiment was performed on a CNC machining center. The workpiece was fixed to a three-component dynamometer, which was attached to the table surface and shared the same direction as the machining center. The arrangement of other sensors is shown in Fig. 8. Furthermore, all device specifications are presented in Table IV. Moreover, the experimental condition parameters are given in Table V. In our experiment, Duracarb APMT 1135 carbide cutters were used as cutting tools. Workpieces had a size of $100 \times 60 \times 60 \text{ mm}^3$ (length $\times$ width $\times$ height). The dry cutting mode was used, i.e., no cutting fluid was used during the process. For the purpose of signal acquisition, the sampling frequency of the acoustic emission sensor was set to 1 MHz, and that of others was set to 10 kHz. After that, we collected monitoring data while processing the workpiece; furthermore, we selected the monitoring data containing abnormal segments for the experimental data. In this case study, three cutting force data sets with anomalies are used to verify the proposed method. These data sets included either multisegment anomalies or one-segment anomaly. The software and hardware configuration used to process these data sets are presented in Table II.

TABLE IV
EXPERIMENTAL BENCH CONFIGURATION

Device	Specification
CNC machining center	DAHENG VMC850
Three-component dynamometer	Kistler 9257B
Triaxial accelerometer	PCB 356A15
Acoustic emission sensor	PAC-WD
Free-field microphone	Brüel & Kjær 4966
Workpiece	Material: Carbon steel C45 Size: $100 \times 60 \times 60 \text{ mm}^3$ (length $\times$ width $\times$ height)
Cutting tool	APMT1135
Charge meter	KISTLER-5015
Data acquisition equipment	DEWE3021

TABLE V
EXPERIMENTAL CONDITION PARAMETER

Parameter	Value
Spindle speed	2500rpm
Feed rate	200mm/min
Back engagement of the cutting edge	2mm
Sampling frequency of acoustic emission sensor	1MHz
Sampling frequency of other sensors	10kHz

B. Multisegments Anomaly Detection Results and Discussion

The two data sets with anomalies—missing or swelling—are shown in Figs. 9(a) and 10(a), respectively. In both cases, raw data window matrices were constructed with a window length of $L_w = 3000$ and a moving step length of $0.1 \times L_w$ to extract features. Feature matrixes were then constructed to train iForest containing 100 iTrees, and calculated anomaly score. Figs. 9(b) and 10(b) show the calculation results of each window. The green point indicates that the abnormal score of the widow exceeds 0.6; hence, it is a PAW. However, it may be a misjudgment to directly classify those windows as abnormal whose anomaly scores exceed 0.6. This is because many windows were normal windows; however, their anomaly scores were larger than or equal to 0.6. Additionally, scores of the abnormal windows inside the abnormal data segment may not be significantly different from the scores of the normal windows, increasing the risk of misjudgment. For instance, anomaly scores of the windows in the red box in Fig. 9(a) are not significantly different from those in the purple box, according to their anomaly scores shown in the same color boxes in Fig. 9(b). Such a situation increases the risk of classifying the windows in the purple box as anomalies. Therefore, a similarity measurement between CSPAW and the CSMPAW is proposed to decrease the risks of misjudgment. The closer the similarity is to 1, the more likely the CSPAW is an abnormal window. Fig. 9(c) shows the similarity between each CSPAW and the CSMPAW. We can see that it is so similar between the continuous sequence of real abnormal windows and the CSMPAW. And the continuous sequence of fake abnormal windows has relatively low similarities with the CSMPAW. Therefore, similarity measurement makes the anomaly detection robust.

The results of multisegment anomaly data detection with a sensitivity of 0.5 are demonstrated in Figs. 9(d) and 10(c), which show that the proposed method can effectively detect anomalies with missing and swelling.

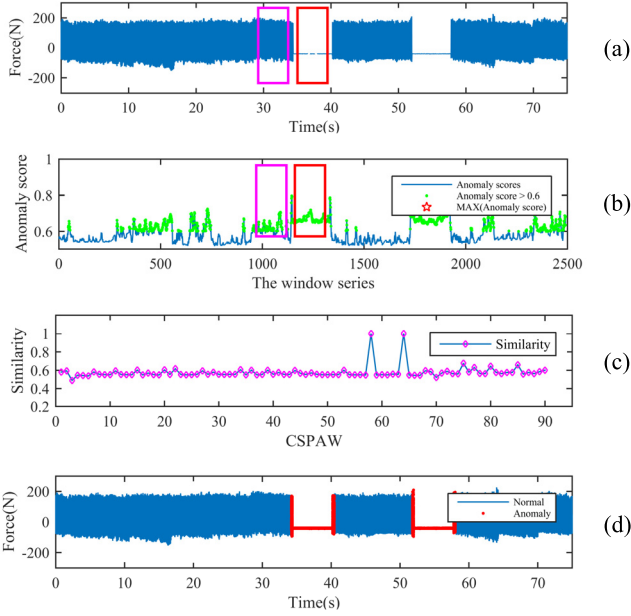


Fig. 9. Multisegment missing data detection. (a) Raw data with two segments missing (RD2M). (b) Abnormal score of each window. (c) Similarity of the CSPAW with the CAMPAW. (d) Detection result of RD2M.

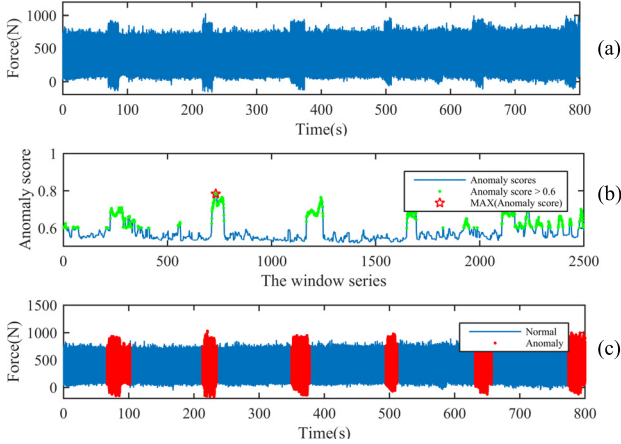


Fig. 10. Multisegment swelling data detection. (a) Raw data with six segments swelling (RD6W). (b) Abnormal score of each window. (c) Detection result of RD6B.

C. One-Segment Anomaly Detection Results and Discussion

The data set containing a shifting segment was used for anomaly detection to further verify SM-iForest, as shown in Fig. 11(b). The parameters except sensitivity are the same as in multisegment anomaly detection.

Similarities between CSPAW and CSMPAW are shown in Fig. 11(a). A point shown in the red circle is observed to be far away from the others, which suggests that one CSPAW has the lowest similarity with the CSMPAW. This point significantly impacts the identification of the abnormal window. Therefore, the proposed method reduces its influence by setting a relative threshold, as in (10), based on the unilateral trimmed mean. Additionally, SM-iForest provides a sensitivity parameter to adjust the relative threshold for accurate detection. The detection results are demonstrated in Fig. 11(b) and (c) with a sensitivity of 0.1 and 0.6, respectively.

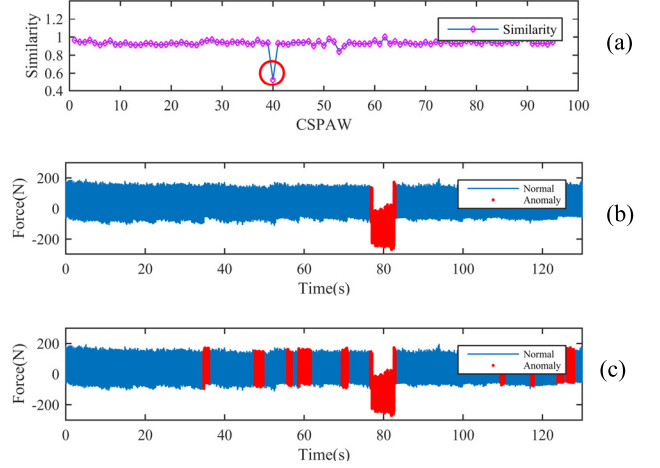


Fig. 11. One-segment shifting data detection. (a) Similarity of the CSPAW with the CAMPAW. (b) Detection result with sensitivity of 0.1. (c) Detection result with sensitivity of 0.6.

It suggests that the proposed method can provide a more controllable detection result to achieve different detection accuracy.

D. Comparative Experiment

1) *Three Baseline Methods:* We compared three baseline methods that are widely used in the field of anomaly detection to further verify the effectiveness of the proposed method. These methods include the original iForest, LOF, and 3σ -rule-based-method.

The raw monitoring data with one-segment anomaly were directly used to construct iForest while using the original iForest to detect MMD anomaly. These data were then fed into iForest to obtain anomaly scores. Finally, data with anomaly scores greater than 0.6 were identified as anomalies. The detection result is given in Fig. 12(a). We observe that the original iForest classified almost all the data as anomalies, which indicates its severe inadaptability when employed directly for MMD anomaly detection. Therefore, we conducted sliding-window processing on raw data before using the original iForest to detect anomalies. We call iForest with this improvement half-improved iForest. The detection result of the half-improved iForest is given in Fig. 12(b), which suggests a conspicuous increase in iForest's adaptability. However, we can still observe from the figure that the false alarm remains relatively high. It implies that the original iForest algorithm is still too sensitive to MMD in distinguishing real anomalies. Therefore, an anomaly identification stage is proposed to solve this problem. The performance of the proposed method is already given in Fig. 11(b), which demonstrates the superiority of SM-iForest.

LOF is a widely used anomaly detection method. It identifies anomalies by calculating the data's LOF. The LOF of the data point, p , is obtained by

$$\text{LOF}_k(p) = \frac{\sum_{o \in N_k(p)} \frac{lrd_k(o)}{lrd_k(p)}}{|N_k(p)|} \quad (16)$$

where $N_k(p)$ denotes the k -distance neighborhood of p , $lrd_k(o)$ shows the local reachability density of point o , $lrd_k(p)$

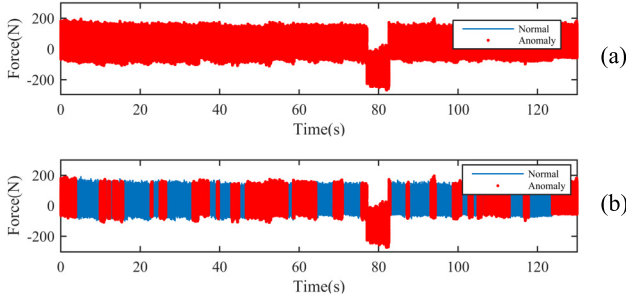


Fig. 12. One-segment anomaly detection result. (a) Detection result of the original iForest algorithm. (b) Detection result of iForest with sliding-window processing.

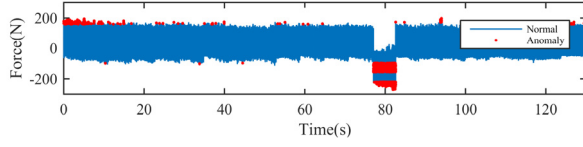


Fig. 13. Detection result of LOF.

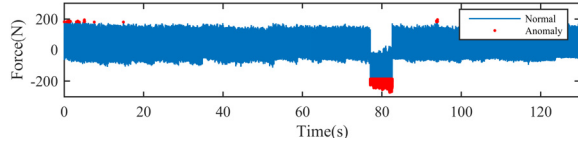


Fig. 14. Detection result of the 3σ -rule-based method.

represents the local reachability density of point p , and $|N_k(p)|$ is the number of points in $N_k(p)$. If the $LOF_k(p)$ is greater than 1, p is an anomaly point. In this detection experiment, the k value is set to 20. Fig. 13 provides detailed detection results using LOF. We notice from the figure the poor detection accuracy of LOF when used for MMD anomaly detection.

The method based on the 3σ rule is a classical anomaly detection method. If random variable, F , obeys normal distribution, $F \sim N(\mu, \sigma^2)$, then its probability density function is

$$g(F) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(F-\mu)^2}{2\sigma^2}} \quad (17)$$

where μ and σ ($\sigma > 0$) are constant. The random variable, F , has only a 0.26% possibility to be located in $(-\infty, \mu - 3\sigma) \cup (\mu + 3\sigma, +\infty)$. Therefore, F might be an anomaly when its value is situated in the interval. The method based on the 3σ rule method was directly used to spot anomalies in the raw data. Its result is shown in Fig. 14. It is found that the method is not effective in detecting anomalies. The result—missing many anomalies—is exactly the opposite of the original iForest.

2) *Four State-of-The-Art Methods*: Additionally, we also compared SM-iForest with four state-of-the-art unsupervised anomaly detection methods: a novel algorithm based on spectral residual and convolutional neural network (SR-CNN) [41], LODA [42], LSCP [43], and copula-based outlier detection (COPOD) [44].

Hansheng *et al.* proposed SR-CNN that introduced the spectral residual model from visual saliency detection to anomaly detection in time series data. Compared with some existing methods, this method has higher detection accuracy. However,

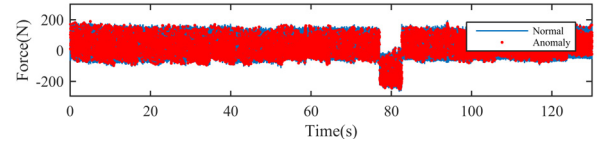


Fig. 15. Detection result of SR-CNN.

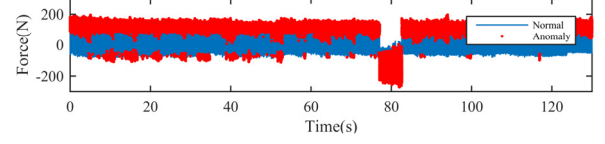


Fig. 16. Detection result of LODA.

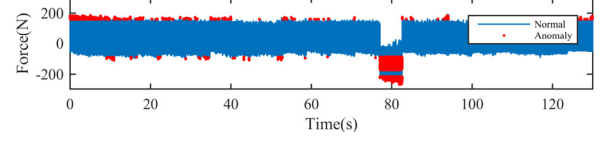


Fig. 17. Detection result of LSCP.

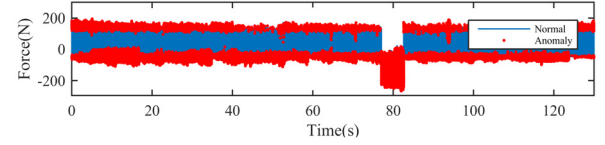


Fig. 18. Detection result of COPOD.

it generated significantly poor outcomes from the detection of MMD anomalies, as shown in Fig. 15.

LODA is considered as one of the representative anomaly detection methods with its overall good performance. Nonetheless, its detection performance on MMD, as shown in Fig. 16, looks not as good as its performance on the datasets used in the literature [42].

LSCP is the state-of-the-art time series data anomaly detection model based on outlier ensembles. In our experiment, the detector list consists of four LOF detectors. The number of neighbors to be used for k -neighbors queries for these LOF detectors are 15, 20, 25, and 35 [45]. The detection performance of LSCP is presented in Fig. 17. It shows that LSCP fails when used for MMD anomaly detection.

Furthermore, COPOD is the latest work on anomaly detection based on probability. The method shows a remarkable detection performance in most of the 30 datasets used in the literature [44]. However, the detection result on MMD indicates that it is not sufficiently effective, as shown in Fig. 18.

3) *Result Comparison of One-Segment Anomaly Detection*: The detection and false alarm rates were used to quantitatively evaluate the comparison of the results of anomaly detection; it was done to show more accurately the detection effectiveness of the proposed method and the comparison methods.

$$P_{\text{detection rate}} = \frac{TP}{TP + FN} \times 100\% \quad (18)$$

$$P_{\text{false alarm rate}} = \frac{FP}{TP + FP} \times 100\%. \quad (19)$$

TP in (18) and (19) is true positive representing the number of anomalies that are classified correctly. FN is a false negative, which indicates the number of anomalies that are

TABLE VI
DETECTION RESULT COMPARISON

Method name	Detection rate	False alarm rate
Original iForest	100%	95.75%
Half-improved iForest	100%	92.09%
SM-iForest	100%	9.20%
LOF	2.80%	14.72%
3 σ -rule-based method	39.21%	0.24%
SR-CNN	1.14%	95.17%
LODA	82.91%	63.41%
LSCP	8.61%	12.11%
COPOD	94.73%	59.79%

not correctly recognized. FP is false positive, representing the number of normal data that are classified as anomalies.

The quantitative comparison results are given in Table VI. It is seen that half-improved iForest demonstrates a decrease in false alarm rate compared to the original iForest. Moreover, the proposed SM-iForest has the lowest false alarm rate among the three. Furthermore, the results show that the SM-iForest has the highest detection rate compared to LOF and method based on the 3 σ rule. Additionally, SM-iForest performs better in detection rates and false alarm rate compared to the four state-of-the-art methods. This is because SM-iForest is specifically designed to detect MMD anomaly. Therefore, the proposed SM-iForest outperforms the three baseline methods and the four state-of-the-art methods in terms of comprehensive performance. It has the highest detection rate and a relatively low false alarm rate.

V. CONCLUSION

In this article, an anomaly detection method—SM-iForest—is proposed to improve the quality of machinery monitoring data. The method utilized sliding-window processing to improve the adaptability of iForest for MMD abnormal data segments detection. The high-performance computing capability of iForest was then used to obtain the anomaly score of each data segment. Finally, the proposed anomaly identification method was used to determine the specific location of anomalies. Based on the experimental results, the following conclusions are drawn.

- 1) SM-iForest overcomes a key shortcoming of the original iForest algorithm by considering an abnormal data cluster as either an abnormal data segment or an abnormal data window. Specifically, original iForest can only deal with the data sets containing a small number of anomalies instead of abnormal data cluster. The segmentation algorithm for raw monitoring data and the feature matrix construction were the concrete measures taken to reach this goal. Moreover, the step to construct the feature matrix also reduces the amount of data processed and improves data processing efficiency.
- 2) SM-iForest improves the detection robustness and corrects a defect of the original iForest by an anomaly identification module, i.e., iForest is unsuitable for processing data with massive irrelevant features.
- 3) This method can detect three of the most common MMD anomalies, including shifting, swelling, and missing

without any prior information. It is suitable for rotating machinery monitoring data anomaly detection.

- 4) The proposed method is more controllable and convenient compared to the state-of-the-art anomaly detection methods. The interface of detection sensitivity is reserved, making it more convenient for readers of both industry and academia in adjusting detection sensitivity to meet their respective detection requirements. Moreover, the proposed method has a considerably lower false alarm rate and higher accuracy than the comparison methods.

In the future, the major goal is to further improve the adaptivity of the proposed method. It would then be possible for MMD anomaly detection to be more accurate and convenient. Furthermore, more efficient and accurate data cleaning methods for anomaly detection and data filling would be developed. The research on data cleaning of MMD is still in its infancy with broad application prospects. Therefore, research on MMD data cleaning is valuable and should be conducted. Another challenging goal is to collect various types of anomalous data to construct a database of “dirty data.” It would bring significant progress in detecting MMD anomaly.

REFERENCES

- [1] W. Ahmad, S. A. Khan, M. M. M. Islam, and J.-M. Kim, “A reliable technique for remaining useful life estimation of rolling element bearings using dynamic regression models,” *Rel. Eng. Syst. Saf.*, vol. 184, pp. 67–76, Apr. 2019.
- [2] A. L. Ellefsen, E. Bjørlykhaug, V. Æsøy, S. Ushakov, and H. Zhang, “Remaining useful life predictions for turbofan engine degradation using semi-supervised deep architecture,” *Rel. Eng. Syst. Saf.*, vol. 183, pp. 240–251, Mar. 2019.
- [3] B. Wang, Y. Lei, N. Li, and N. Li, “A hybrid prognostics approach for estimating remaining useful life of rolling element bearings,” *IEEE Trans. Rel.*, vol. 69, no. 1, pp. 401–412, Mar. 2020.
- [4] L. Guo, Y. Lei, S. Xing, T. Yan, and N. Li, “Deep convolutional transfer learning network: A new method for intelligent fault diagnosis of machines with unlabeled data,” *IEEE Trans. Ind. Electron.*, vol. 66, no. 9, pp. 7316–7325, Sep. 2019.
- [5] D. Goyal, A. Choudhary, B. S. Pabla, and S. S. Dhami, “Support vector machines based non-contact fault diagnosis system for bearings,” *J. Intell. Manuf.*, vol. 31, no. 5, pp. 1275–1289, Jun. 2020.
- [6] Z. You, H. Gao, L. Guo, Y. Liu, and J. Li, “On-line milling cutter wear monitoring in a wide field-of-view camera,” *Wear*, vols. 460–461, Nov. 2020, Art. no. 203479.
- [7] Z. Huo, M. Martinez-Garcia, Y. Zhang, R. Yan, and L. Shu, “Entropy measures in machine fault diagnosis: Insights and applications,” *IEEE Trans. Instrum. Meas.*, vol. 69, no. 6, pp. 2607–2620, Jun. 2020.
- [8] Z. Chen and W. Li, “Multisensor feature fusion for bearing fault diagnosis using sparse autoencoder and deep belief network,” *IEEE Trans. Instrum. Meas.*, vol. 66, no. 7, pp. 1693–1702, Jul. 2017.
- [9] H. Chen and B. Jiang, “A review of fault detection and diagnosis for the traction system in high-speed trains,” *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 2, pp. 450–465, Feb. 2020.
- [10] I. Taleb, R. Dssouli, and M. A. Serhani, “Big data pre-processing: A quality framework,” in *Proc. IEEE Int. Congr. Big Data*, Jun. 2015, pp. 191–198.
- [11] M. M. Najafabadi, F. Villanustre, T. M. Khoshgoftaar, N. Seliya, R. Wald, and E. Muharemagic, “Deep learning applications and challenges in big data analytics,” *J. Big Data*, vol. 2, no. 1, pp. 1–21, Dec. 2015.
- [12] L. Guo, Y. Lei, N. Li, T. Yan, and N. Li, “Machinery health indicator construction based on convolutional neural networks considering trend burr,” *Neurocomputing*, vol. 292, pp. 142–150, May 2018.
- [13] H. Chen, B. Jiang, W. Chen, and H. Yi, “Data-driven detection and diagnosis of incipient faults in electrical drives of high-speed trains,” *IEEE Trans. Ind. Electron.*, vol. 66, no. 6, pp. 4716–4725, Jun. 2019.

- [14] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Comput. Surv.*, vol. 41, no. 3, pp. 1–58, Jul. 2009.
- [15] W. Wang, D. Lu, X. Zhou, B. Zhang, and J. Mu, "Statistical wavelet-based anomaly detection in big data with compressive sensing," *EURASIP J. Wireless Commun. Netw.*, vol. 2013, no. 1, pp. 1–6, Dec. 2013.
- [16] P. Liang, C. Deng, J. Wu, G. Li, Z. Yang, and Y. Wang, "Intelligent fault diagnosis via semisupervised generative adversarial nets and wavelet transform," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 7, pp. 4659–4671, Jul. 2020.
- [17] H. Chen, B. Jiang, S. X. Ding, N. Lu, and W. Chen, "Probability-relevant incipient fault detection and diagnosis methodology with applications to electric drive systems," *IEEE Trans. Control Syst. Technol.*, vol. 27, no. 6, pp. 2766–2773, Nov. 2019.
- [18] S. Hawkins, H. He, G. Williams, and R. Baxter, "Outlier detection using replicator neural networks," in *Proc. Int. Conf. Data Warehousing Knowl. Discovery*, 2002, pp. 170–180.
- [19] X. Yang, L. J. Latecki, and D. Pokrajac, "Outlier detection with globally optimal exemplar-based GMM," in *Proc. SIAM Int. Conf. Data Mining*, Apr. 2009, pp. 145–154.
- [20] S. S. Khan and B. Taati, "Detecting unseen falls from wearable devices using channel-wise ensemble of autoencoders," *Expert Syst. Appl.*, vol. 87, pp. 280–290, Nov. 2017.
- [21] D. M. J. Tax and R. P. W. Duin, "Support vector domain description," *Pattern Recognit. Lett.*, vol. 20, nos. 11–13, pp. 1191–1199, Nov. 1999.
- [22] J. Hardin and D. M. Rocke, "Outlier detection in the multiple cluster setting using the minimum covariance determinant estimator," *Comput. Statist. Data Anal.*, vol. 44, no. 4, pp. 625–638, Jan. 2004.
- [23] F. E. Grubbs, "Procedures for detecting outlying observations in samples," *Technometrics*, vol. 11, no. 1, pp. 1–21, Feb. 1969.
- [24] Z. Bakar, R. Mohamad, A. Ahmad, and M. Deris, "A comparative study for outlier detection techniques in data mining," in *Proc. IEEE Conf. Cybern. Intell. Syst.*, Jun. 2006, pp. 1–6.
- [25] M. L. Shyu, S. C. Chen, K. Sarinapakorn, and L. Chang, "A novel anomaly detection scheme based on principal component classifier," in *Proc. IEEE Found. New Directions Data Mining*, Jan. 2003, pp. 171–179.
- [26] A. Lakhina, M. Crovella, and C. Diot, "Mining anomalies using traffic feature distributions," in *Proc. Conf. Appl., Technol., Archit., Protocols Comput. Commun. SIGCOMM*, 2005, pp. 217–228.
- [27] K.-V. Yuen and G. A. Ortiz, "Outlier detection and robust regression for correlated data," *Comput. Methods Appl. Mech. Eng.*, vol. 313, pp. 632–646, Jan. 2017.
- [28] E. M. Jordaán and Z. F. Smits, "Robust outlier detection using SVM regression," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, vol. 3, Jul. 2004, pp. 2017–2022.
- [29] A. F. Emmott, S. Das, T. Dietterich, A. Fern, and W.-K. Wong, "Systematic construction of anomaly detection benchmarks from real data," in *Proc. ACM SIGKDD Workshop Outlier Detection Description ODD*, 2013, pp. 16–21.
- [30] V. Hautamaki, I. Karkkainen, and P. Franti, "Outlier detection using k-nearest neighbour graph," in *Proc. 17th Int. Conf. Pattern Recognit. ICPR*, Aug. 2004, pp. 430–433.
- [31] M. M. Breunig, H. P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2000, pp. 93–104.
- [32] X. Xu, Y. Lei, and Z. Li, "An incorrect data detection method for big data cleaning of machinery condition monitoring," *IEEE Trans. Ind. Electron.*, vol. 67, no. 3, pp. 2326–2336, Mar. 2020.
- [33] F. T. Liu, K. M. Ting, and Z. H. Zhou, "Isolation forest," in *Proc. IEEE Int. Conf. Data Mining*, Dec. 2008, pp. 413–422.
- [34] J. Söpanen and A. Mikkola, "Dynamic model of a deep-groove ball bearing including localized and distributed defects. Part 1: Theory," *Proc. Inst. Mech. Eng., K, J. Multi-body Dyn.*, vol. 217, no. 3, pp. 201–211, Sep. 2003.
- [35] A. K. Elmagarmid, P. G. Ipeirotis, and V. S. Verykios, "Duplicate record detection: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 19, no. 1, pp. 1–16, Jan. 2007.
- [36] Z. Zou, Y. Xie, K. Huang, G. Xu, D. Feng, and D. Long, "A docker container anomaly monitoring system based on optimized isolation forest," *IEEE Trans. Cloud Comput.*, early access, Aug. 20, 2019, doi: 10.1109/TCC.2019.2935724.
- [37] X. Tao, Y. Peng, F. Zhao, P. Zhao, and Y. Wang, "A parallel algorithm for network traffic anomaly detection based on isolation forest," *Int. J. Distrib. Sensor Netw.*, vol. 14, no. 11, Nov. 2018, Art. no. 155014771881447.
- [38] H. Chen, B. Jiang, N. Lu, and Z. Mao, "Deep PCA based real-time incipient fault detection and diagnosis methodology for electrical drive in high-speed trains," *IEEE Trans. Veh. Technol.*, vol. 67, no. 6, pp. 4819–4830, Jun. 2018.
- [39] H. Wang, S. Li, L. Song, L. Cui, and P. Wang, "An enhanced intelligent diagnosis method based on multi-sensor image fusion via improved deep learning network," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 6, pp. 2648–2657, Jun. 2020.
- [40] D. Ho and R. B. Randall, "Optimisation of bearing diagnostic techniques using simulated and actual bearing fault signals," *Mech. Syst. Signal Process.*, vol. 14, no. 5, pp. 763–788, Sep. 2000.
- [41] H. Ren et al., "Time-series anomaly detection service at microsoft," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2019, pp. 3009–3017.
- [42] T. Pevný, "Loda: Lightweight on-line detector of anomalies," *Mach. Learn.*, vol. 102, no. 2, pp. 275–304, Feb. 2016.
- [43] Y. Zhao, Z. Nasrullah, M. K. Hryniewicki, and Z. Li, "LSCP: Locally selective combination in parallel outlier ensembles," in *Proc. SIAM Int. Conf. Data Mining*, 2019, pp. 585–593.
- [44] Z. Li, Y. Zhao, N. Botta, C. Ionescu, and X. Hu, "COPD: Copula-based outlier detection," 2020, *arXiv:2009.09463*. [Online]. Available: <http://arxiv.org/abs/2009.09463>
- [45] Y. Zhao, Z. Nasrullah, and Z. Li, "PyOD: A Python toolbox for scalable outlier detection," *J. Mach. Learn. Res.*, vol. 20, no. 96, pp. 1–7, Jan. 2019.



Changgen Li was born in Guangyuan, China. He received the B.S. degree in mechatronic engineering from the Xihua University, Chengdu, China, in 2018. He is currently pursuing the Ph.D. degree with the Southwest Jiaotong University, Chengdu. His research interests include condition monitoring, anomaly detection, and data cleaning.



Liang Guo received the B.S. and Ph.D. degrees in mechanical engineering from the Southwest Jiaotong University, Chengdu, China, in 2011 and 2016, respectively.

He was a Postdoctoral Research Fellow with the Xi'an Jiaotong University, Xi'an, China, in 2018. He was also a Visiting Ph.D. Student with the Clemson University, Clemson, SC, USA. He is currently an Assistant Professor with Southwest Jiaotong University. His research interests include machinery intelligent fault diagnostics and remaining useful life prediction.



Hongli Gao received the Ph.D. degree in mechanical engineering from the Southwest Jiaotong University, Chengdu, China, in 2005.

He is currently a Professor with the Southwest Jiaotong University. His research interests include designing and reliability analysis of complex electromechanical equipment systems, intelligent machinery state monitoring, and fault diagnosis.



Yi Li received the B.S. degree in mechatronic engineering from the Xihua University, Chengdu, China, in 2017. She is currently pursuing the M.S. degree in mechatronic engineering with the Southwest Jiaotong University, Chengdu, China.

Her current research interests include signal processing, tool wear condition monitoring, and prognostics health management.